

Signal Geometry and Fusion Order in Multiview Entity Resolution:

A 2×2 Analysis on CORA and FEBRL1

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Abstract

Multiview entity resolution (ER) combines several partial views of a record (e.g. name, address, bibliographic title) to decide which records refer to the same entity. The literature offers many fusion architectures but little guidance on *when* fusion helps. We organize the design space into a 2×2 taxonomy with two axes: the **representation space** in which views are combined (pairwise *similarity* matrices vs. *graphs*) and the **strategy order** (*fuse-then-cluster* vs. *cluster-then-consensus*), giving four method families: A-Sim, B-Sim, A-Graph, and B-Graph. We complete the grid’s fourth cell, B-Graph (graph-clustering-then-consensus), and benchmark all four cells under one frozen protocol on two single-source deduplication datasets — CORA (real bibliographic) and FEBRL1 (synthetic person records). Three findings emerge. (i) *Representation space dominates*: on both datasets the best similarity-space method beats the best graph-space method. (ii) *Order matters only in graph space*: A-Sim and B-Sim are near-identical, but A-Graph and B-Graph differ by up to $12\times$ and the better order flips between datasets; in particular, cluster-first ordering rescues FEBRL1 graph ER from 0.06 to 0.74 ARI by averting giant-component percolation. (iii) Whether multiview fusion beats the single best view is governed by a *Complementary Gain Index* (CGI): $\text{CGI} < 1$ (CORA, concentrated signal) means fusion hurts, $\text{CGI} > 1$ (FEBRL1, distributed signal) means it helps; a naive single-view dominance index (SCI) does *not* discriminate. We also de-confound the weighting $\times$ threshold axis, showing equal weighting is best or tied everywhere. Every number is tied to a reproducible run. We frame all geometry claims as two-dataset evidence, note that the representation axis is confounded with the clustering algorithm, and identify a third real dataset as the principal step toward generality.

1 Introduction

Entity resolution (ER) — deciding which records refer to the same real-world entity — underpins data integration, deduplication, and record linkage [4, 5]. Modern records are *multiview*: a person record exposes name, address, and locality fields; a citation exposes title, authors, and venue. A long line of work fuses such views, via multi-kernel learning [11], similarity network fusion [22], multiplex community detection [13], and consensus clustering [8, 17]. Yet practitioners lack a simple answer to a basic question: *given a new dataset, will multiview fusion help, and which architecture should I use?*

The field’s reflex is to compare named algorithms (“method A beats method B”). We argue this reductionism hides the real structure. We recast the design space as a 2×2 grid along two axes:

- **Representation space**: are views combined as pairwise *similarity* matrices, or as *graphs* (using topology — neighborhoods, density, diffusion)?

- **Strategy order:** *fuse-then-cluster* (combine all views, cluster once) vs. *cluster-then-consensus* (cluster each view, then reconcile).

The four cells are A-Sim, B-Sim, A-Graph, B-Graph. We implement all four under one harness and run the full grid on CORA and FEBRL1.

Contributions. (1) A 2×2 taxonomy that separates two effects the literature conflates — representation space and fusion order (§4). (2) A uniform implementation of all four cells under one protocol, including the cluster-first graph cell B-Graph (§5). (3) A two-dataset empirical study showing similarity space dominates graph space, order matters only in graph space, and a quantitative geometry index (CGI) predicts whether fusion beats the best single view (§7–§8). (4) A de-confounded weighting analysis and an honest claim ledger scoping every result to its evidence.

2 Related Work

Entity resolution. Foundations include the Fellegi–Sunter probabilistic model [7], string comparators [20, 23], collective/relational ER [1], the sorted-neighborhood method [9], blocking surveys [15], and deep entity matching [12, 14]. **Multiview / multi-kernel learning.** Sun [18] surveys multiview learning; Lanckriet et al. [11] learn convex kernel combinations — the conceptual root of similarity fusion (our A-Sim). **Graph fusion.** Similarity Network Fusion [22] and multiplex community detection [13] fuse per-view graphs (our A-Graph), clustered with Louvain [2] or Leiden [19] on *igraph* [6]. **Consensus clustering.** Cluster ensembles [17] and evidence accumulation [8] combine partitions (our B-Sim / B-Graph consensus step). **Evaluation.** We use the adjusted Rand index [10] as the primary metric, with NMI / V-measure [16, 21] and pairwise F1 secondary. We organize these threads into a single representation $\times$ order grid, so that the two effects they usually vary together — representation space and fusion order — can be read off independently.

3 Problem Setting

A deduplication instance is a record set $R = \{r_1, \dots, r_n\}$ with a set of views $V = \{v_1, \dots, v_m\}$ (disjoint field groups). For a candidate-pair set P (here all pairs; blocking = none), each view v induces a pairwise similarity $S_v(i, j) \in [0, 1]$ and, equivalently, a weighted graph $G_v = (R, E_v, S_v)$. The task is to output a partition of R into predicted entities, scored against ground truth by ARI (primary), NMI, VI, $\text{FMI} = \sqrt{\text{PairP} \cdot \text{PairR}}$, and pairwise F1. We restrict attention to single-source deduplication so that pair generation is held fixed across all methods.

4 A 2×2 Taxonomy of Multiview ER

Table 1 and Figure 1 define the grid.

Dimension 1 — representation space. *Similarity space* operates on the weights S_v with an absolute threshold or classifier. *Graph space* operates on G_v as a graph, using community detection, network fusion, or diffusion that exploit topology, not only an absolute cutoff. The two share one underlying object — a similarity matrix S_v is exactly the weighted adjacency of G_v — so the axis distinguishes not the data structure but the *edge-selection operation*: gating candidate links by an absolute similarity cutoff (similarity space) versus topology-driven clustering with no absolute cutoff

Table 1: The 2×2 taxonomy of multiview ER. Rows = representation space; columns = strategy order. All four cells are implemented and run under one protocol.

	A: fuse → cluster	B: cluster → consensus
Similarity	A-Sim: combine $S = \sum_v w_v S_v$, threshold, connected components	B-Sim: threshold-cluster each view, co-membership consensus at α
Graph	A-Graph: fuse graphs (SNF / multiplex), then community detection	B-Graph: community-detect each view graph, consensus at α

(graph space). This operational reading is what the empirical gap turns on (§8), and it also explains why the axis is entangled with the choice of clustering algorithm (§9).

Dimension 2 — strategy order. *Strategy A* fuses all views into one object, then clusters once. *Strategy B* clusters each view independently, then reconciles via co-membership consensus at a fraction α of views.

5 Methods

All four cells reuse one ER toolkit (real implementations; no method-specific tuning beyond the fixed per-cell grids of §6).

A-Sim. Fused similarity $S(i, j) = \sum_v w_v S_v(i, j)$ with NA-aware weights w_v (equal, variance, or Fellegi–Sunter). Edges are pairs with $S \geq \tau$; entities are connected components. Knobs: weighting, threshold τ .

B-Sim. For each view, form components at threshold τ . A co-membership counter over pairs retains an edge iff at least $\lceil \alpha m \rceil$ views co-cluster it; entities are components of the consensus graph [8, 17]. Knobs: τ, α .

A-Graph. Build per-view graphs and fuse either by SNF [22] (cross-normalized diffusion, K neighbors, t iterations) or by a multiplex supra-adjacency [13] (coupling ω , resolution γ); cluster the fused graph with Louvain [2]. Knobs: graph construction (dense / KNN- K), K, t, γ, ω .

B-Graph. For each view graph G_v (dense or KNN-sparsified), run Louvain to get a per-view partition C_v ; combine $\{C_v\}$ by the same co-membership consensus as B-Sim (fraction α); entities are components of the consensus graph. Clustering each view graph *before* combining prevents cross-view noise edges from chaining into one giant component prior to clustering — the mechanism behind A-Graph’s collapse (§8). Knobs: construction (dense / KNN- K), α .

A 2×2 taxonomy of multiview ER

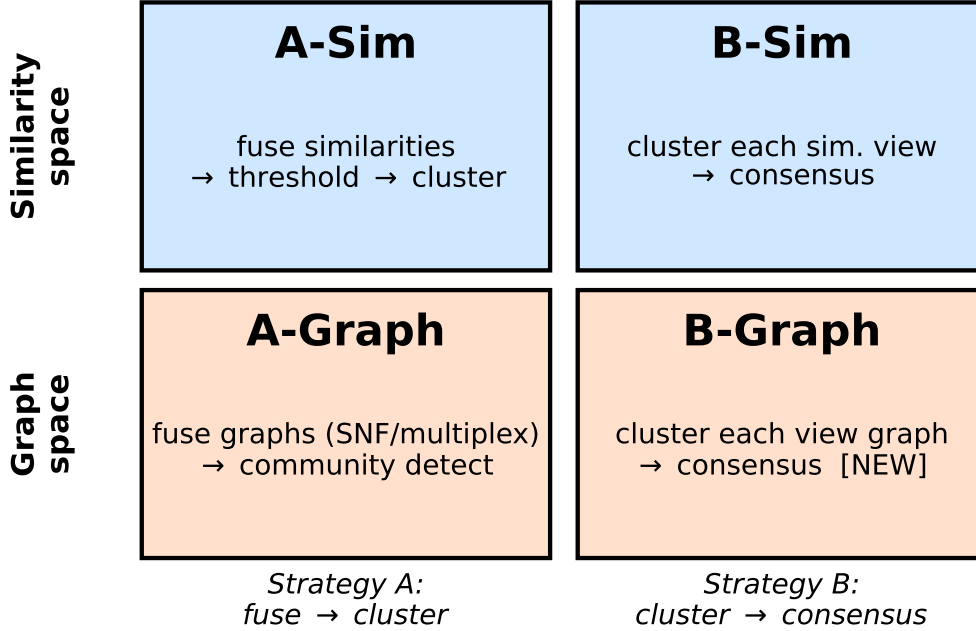


Figure 1: The 2×2 taxonomy. Rows: representation space (similarity vs. graph). Columns: strategy order (fuse-then-cluster vs. cluster-then-consensus).

6 Experimental Design

Datasets and field selection. CORA (1,879 bibliographic records, 118 labeled clusters) has 15 raw fields, most of them sparsely populated (e.g. `institution` 13%, `note` 4%). We form three *atomic* single-field views from its usable fields — `title` (98% populated) and `authors` (99.8%), the two densest and most discriminative, plus `address` — matching FEBRL1’s view count for a balanced fusion comparison; token-Jaccard similarity. FEBRL1 (1,000 synthetic person records, 500 size-2 clusters) is dataset 1 of the FEBRL record-linkage benchmark [3], generated as 500 originals each paired with a single, lightly modified duplicate. Of its 10 fields we exclude two near-unique identifiers (leakage controls, below) and group the remaining eight into three semantic *grouped* views — `name` (given/surname), `address` (street number/lines), `geo` (suburb/postcode/state); Jaro-Winkler + categorical similarity. Both are single-source deduplication, so pair generation is held fixed across all methods. The two datasets thus differ in view *granularity* (atomic vs. grouped), an intrinsic property we flag in §9.

Protocol parity. One frozen protocol: blocking none (all pairs), `labeled_only` evaluation, identical metrics, identical CSV schema, seed 42. The only cross-dataset differences are intrinsic field properties (token-Jaccard vs. Jaro-Winkler; partial vs. full labeling), documented as such.

Table 2: Dataset summary. Both are single-source deduplication corpora evaluated under one frozen protocol (blocking none, `labeled_only` evaluation, seed 42).

Property	CORA	FEBRL1	Notes
Records / truth clusters	1,879 / 118	1,000 / 500	CORA partially labeled
Type	bibliographic	synthetic person	real vs. synthetic
Views	title, authors, address	name, address, geo	3 views each
Similarity	token-Jaccard	Jaro-Winkler + categorical	field-type intrinsic
Geometry (CGI)	concentrated (0.82)	distributed (1.18)	see Table 4
Excluded (leakage)	—	<code>soc_sec_id</code> , <code>date_of_birth</code>	near-unique IDs

Table 3: Best ARI per 2×2 cell (with NMI, VI, FMI, PairF1) for each dataset. Run identifiers index `multiview_2x2_results.csv`. **Bold** = best cell on that dataset. The single-view reference is the best individual view.

Data	Cell	ARI	NMI	VI	FMI	PairF1	run
CORA	single (title)	0.8665	0.952	0.375	0.874	0.872	X002
CORA	A-Sim	0.7025	0.884	0.921	0.721	0.716	X012
CORA	B-Sim	0.7069	0.901	0.765	0.734	0.721	X032
CORA	A-Graph	0.6286	0.815	1.267	0.675	0.647	X041
CORA	B-Graph	0.3061	0.822	1.364	0.365	0.343	X059
FEBRL1	single (geo)	0.8447	0.985	0.185	0.852	0.845	X075
FEBRL1	A-Sim	0.9869	0.999	0.015	0.987	0.987	X076
FEBRL1	B-Sim	0.9990	1.000	0.001	0.999	0.999	X101
FEBRL1	A-Graph	0.0602	0.732	2.688	0.161	0.062	X111
FEBRL1	B-Graph	0.7353	0.977	0.296	0.742	0.736	X127

Leakage controls. FEBRL1’s `soc_sec_id` and `date_of_birth` are near-unique identifiers: including them would reduce multiview ER to exact primary-key matching and trivialize the fusion question. We *exclude* them from all multiview comparisons (they appear only as leakage baselines, exact-match ARI 0.947 / 0.925), preserving the non-triviality of the study.

Indices. For per-view best ARI a_v and best fused ARI a_* , the *Complementary Gain Index* is $\text{CGI} = a_*/\max_v a_v$ (fusion helps iff $\text{CGI}>1$), and the *Signal Concentration Index* is $\text{SCI} = \max_v a_v/\overline{a_v}$.

7 Results

The grid comprises 132 runs (X001–X132). Table 3 gives the best configuration per cell; Figure 2 visualizes it.

Dataset-level. On CORA no multiview cell beats the single best view (title 0.8665); the best multiview is B-Sim 0.7069. On FEBRL1 fusion is decisive: B-Sim reaches 0.9990, far above the best

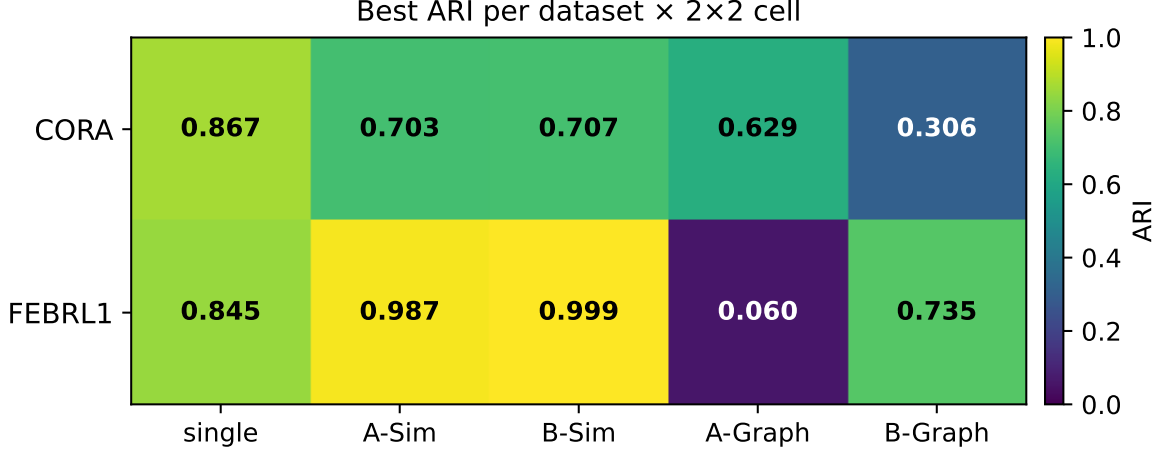


Figure 2: Best ARI per dataset $\times$ cell. Similarity-space cells (A-Sim, B-Sim) dominate graph-space cells (A-Graph, B-Graph) on both datasets.

single view (geo 0.8447).

The 2×2 contrast. Table 4 aligns the two datasets. On both, $\max(\text{similarity cell}) > \max(\text{graph cell})$. Within similarity space the order effect is tiny (≤ 0.012). Within graph space it is large and *flips*: A-Graph beats B-Graph on CORA (0.629 vs. 0.306) but B-Graph crushes A-Graph on FEBRL1 (0.735 vs. 0.060).

Robustness: weighting de-confounding. Figure 3 sweeps weighting $\times$ threshold for A-Sim. Equal weighting is best or tied at every threshold on both datasets; adaptive weightings (variance, Fellegi–Sunter) are neutral on CORA and *catastrophic* on FEBRL1 across the whole neighborhood ($\text{ARI} \approx 0.006$), because they down-weight the discriminative geo view and leave sub-threshold noise that percolates. The weighting effect is thus not a single-threshold artifact.

8 Analysis

Signal geometry. Figure 4 and Table 4 show why the datasets disagree. CORA has $\text{CGI} = 0.816 < 1$: one view (title) is near-sufficient and fusion dilutes it (*concentrated*). FEBRL1 has $\text{CGI} = 1.183 > 1$: no view suffices and fusion aggregates complementary evidence (*distributed*). Crucially, SCI is *not* discriminative (1.625 vs. 1.538), and mean inter-view partition agreement is near-equal (0.267 vs. 0.275): single-view dominance does not imply sufficiency. CGI — the complementary *gain*, not the peakedness of competence — is the operative quantity. We therefore advance CGI as a quantity that motivates, but on two datasets does not prove, an a-priori prediction of fusion benefit.

Why fusion helps or hurts. Within similarity space, both orderings select edges by a high *absolute* threshold, so they are robust to per-view quality and win or lose only by whether the *combined* signal is concentrated or distributed — exactly what CGI measures. This explains the near-identity of A-Sim and B-Sim.

Table 4: CORA vs. FEBRL1 along the two design axes, and signal-geometry indices. $\Delta(A,B)$ is the similarity-space ordering effect; the graph-space ordering effect is large and flips sign. CGI = best-fused ARI / best single-view ARI.

Dimension	CORA	FEBRL1
Best single view	0.8665 (title)	0.8447 (geo)
Max similarity-space cell	0.7069	0.9990
Max graph-space cell	0.6286	0.7353
Sim. > graph?	yes (0.707 > 0.629)	yes (0.999 >> 0.735)
$\Delta(A\text{-Sim}, B\text{-Sim})$	+0.004 (n.s.)	+0.012
A-Graph vs. B-Graph	0.629 > 0.306	0.060 << 0.735
Multiview vs. single	hurts	helps
max-view ARI	0.8665	0.8447
mean-view ARI	0.5332	0.5492
best-fused ARI	0.7069	0.9990
CGI	0.816 (< 1)	1.183 (> 1)
SCI (non-discriminative)	1.625	1.538
mean inter-view ARI	0.267	0.275
Geometry	concentrated	distributed

Why graph methods fail or succeed. The pathological performance of A-Graph on FEBRL1 is structural rather than stochastic. It stems from the topological vulnerability of density-based multiplex / diffusion clustering when operating on non-discriminative Jaro–Winkler affinity matrices, which trigger catastrophic giant-component percolation: because graph space applies *no* absolute cutoff, fusing graphs *first* lets moderate-similarity noise edges from different views chain into one community before any clustering occurs. B-Graph averts this by clustering each view graph *first* (lifting FEBRL1 from 0.060 to 0.735), but still trails similarity space because community detection on noisy per-view graphs over-merges. On CORA, where token-Jaccard is discriminative, fuse-first graph fusion (A-Graph 0.629) beats cluster-first (B-Graph 0.306). Hence graph-method failure is a property of *ordering and construction on weak affinities*, not of graph representation universally.

9 Limitations

- **Two datasets.** The geometry thesis rests on one concentrated and one distributed corpus; it is a two-point contrast. A third real dataset (e.g. NCVr voter registration, plausibly *intermediate* geometry) is the principal step toward generality and is not run here.
- **Representation is confounded with the clustering algorithm.** Similarity cells use threshold+connected-components while graph cells use Louvain, and the two are never crossed. The similarity-vs-graph gap therefore cannot be cleanly separated from a threshold-CC-vs-Louvain gap. Consistent with the *absence of an absolute threshold* — rather than the graph representation as such — driving the collapse, applying Louvain to the fused similarity matrix in companion protocol runs also fails. Crossing controls (Louvain on the thresholded graph; connected components on the KNN graph) are needed to identify the effect and are not run here.
- **CORA’s geometry is partly a field-selection artifact.** CORA’s three views are single fields, one of which (**address**) is sparse (39% populated) and weak in isolation (ARI 0.11); a stronger third view (e.g. venue or year) could redistribute per-view competence and soften the “concentrated” label. Alternative CORA view sets were not tested.
- **“View” differs in granularity across datasets:** atomic single fields on CORA versus

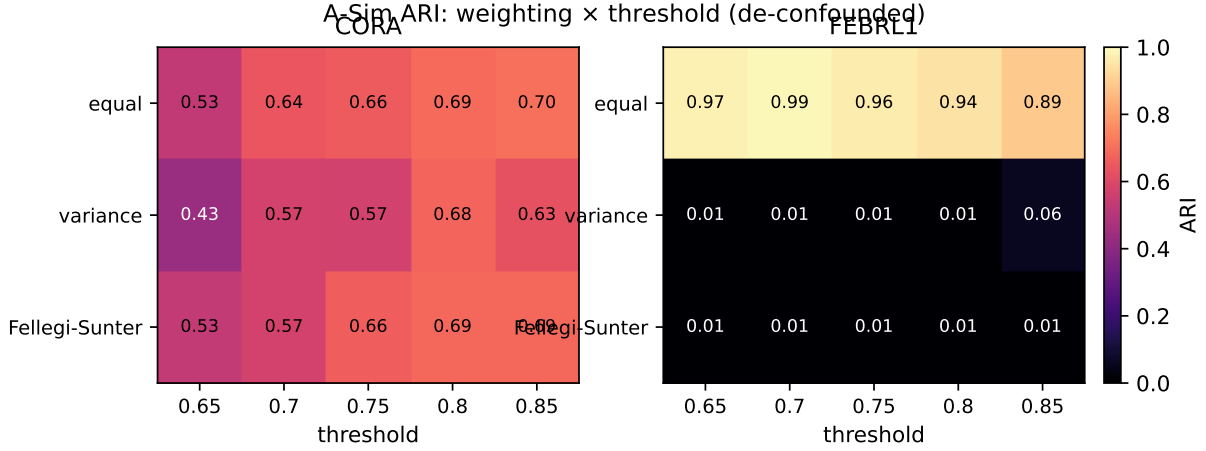


Figure 3: A-Sim ARI over weighting × threshold. Equal weighting dominates; adaptive weighting collapses on FEBRL1 across all thresholds (de-confounded).

grouped multi-field bundles on FEBRL1.

- **Single-source dedup only.** Two-source record linkage (bipartite matching) changes pair generation and is untested; we conjecture but do not show the framework transfers.
- **FEBRL1 is synthetic** (size-2 clusters); CORA is real but small and partially labeled.
- **Graph constructions tested** are dense and KNN- K with Louvain; other affinities, sparsifiers, or detectors could shift the graph-space results.
- **CGI / SCI** are descriptive indices computed post hoc from best-threshold ARI; we do not claim they predict out-of-sample without the third dataset.

10 Conclusion

Reframing multiview ER as a 2×2 grid of representation space × fusion order yields a cleaner account than algorithm-vs-algorithm comparison. On two single-source datasets: representation space sets the achievable ceiling (similarity $\gg$ graph); a complementary-gain index (CGI) governs whether fusion beats the best single view; and ordering matters only inside the fragile graph space, where cluster-first ordering (B-Graph) averts the giant-component collapse that sinks fuse-first graph fusion. The immediate next steps are a third real dataset to test whether CGI predicts the fusion regime a priori, a de-confounding cross of clustering algorithm against representation space to sharpen the similarity-vs-graph claim, and an extension to two-source linkage. Every result here is tied to a reproducible run identifier; claims are scoped to the two-dataset evidence that supports them (Table 5).

Reproducibility and Provenance

All numbers index `results/multiview_2x2_20260623_0900/multiview_2x2_results.csv` (132 runs, X001–X132) and `signal_geometry_indices.csv`, produced by `_scripts/run_2x2.R` (seed 42, R 4.5.0). Per-cell bests reproduce the prior protocol-v2 runs to within rounding. Metrics via the toolkit’s evaluator (mclust / aricode). Graph cells use Louvain community detection on `igraph`; a Leiden variant was attempted but did not complete on these graphs and is not reported.

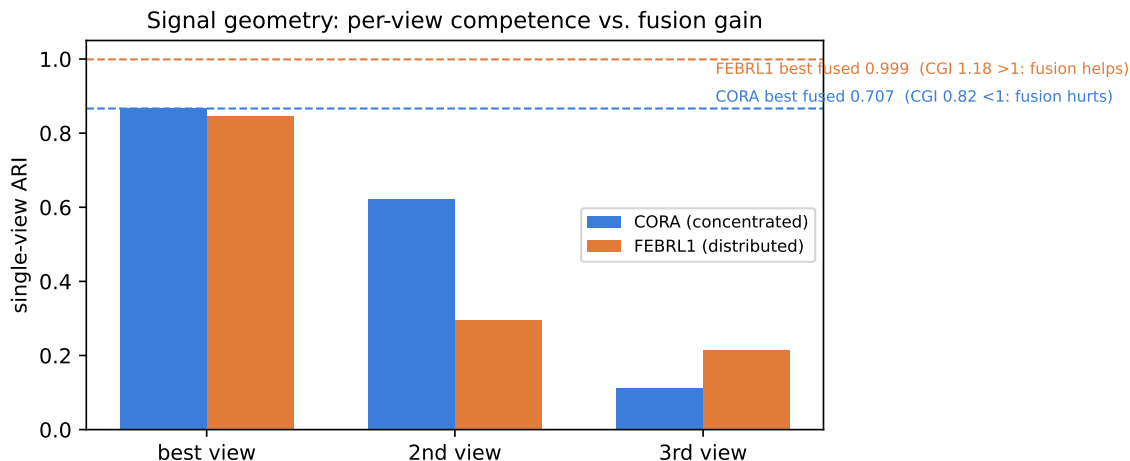


Figure 4: Per-view single-view ARI (sorted). CORA’s competence is concentrated in one view yet fusion hurts ($\text{CGI} < 1$); FEBRL1’s is spread and fusion helps ($\text{CGI} > 1$).

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Table 5: Summary of hypotheses and empirical support. Support level: *verified* (two-dataset, run-backed); *dataset-specific*; *tentative*; *unsupported*. Run identifiers index `multiview_2x2_results.csv`.

Hypothesis	Evidence	Support
Similarity-space cells dominate graph-space cells on both datasets.	X012/X032>X041/X059 X076/X101>X111/X127	verified
Strategy order (A vs. B) is negligible in similarity space.	$\Delta \leq 0.012$ (X012/X032; X076/X101)	verified
Strategy order is decisive in graph space and flips by dataset.	CORA A-Graph>B- Graph; FEBRL1 B- Graph $\gg$ A-Graph	verified
CGI predicts the sign of the fusion benefit (CGI<1 hurts, >1 helps).	CORA 0.82 / FEBRL1 1.18	dataset-specific
A-Graph’s FEBRL1 collapse is structural (giant-component percolation) and mitigated by cluster-first ordering.	X103–X120 vs. X127	verified
Equal weighting $\geq$ adaptive weighting; adaptive catastrophic on FEBRL1.	weighting $\times$ threshold grid	verified (dataset- specific)
Signal geometry <i>determines</i> architecture effectiveness in general.	only 2 datasets	unsupported (pending 3rd)
SCI (single-view dominance) predicts geometry.	1.625 vs. 1.538, non- discriminative	unsupported (re- jected)

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